Machine Learning in Human Motion Analysis HW 3

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1 Derivation of the Principal Component Analysis

The data matrix is:

$$X = \begin{bmatrix} 0 & 4 \\ 1 & 3 \\ 2 & 2 \\ 2 & 5 \\ 3 & 4 \\ 4 & 6 \end{bmatrix}$$

First, we center the data by subtracting the mean of each column:

$$\bar{X} = \begin{bmatrix} -2 & 0 \\ -1 & -1 \\ 0 & -2 \\ 0 & 1 \\ 1 & 0 \\ 2 & 2 \end{bmatrix}$$

Second, we compute the covariance matrix:

$$C = \bar{X}^T \bar{X} = \begin{bmatrix} 10 & 5 \\ 5 & 10 \end{bmatrix}$$

Last, we compute the eigenvalues and their corresponding eigenvectors of the covariance matrix.

We have

$$CX = \lambda X, \quad C - \lambda I = O, \quad \det(C - \lambda I) = 0$$

Solve the characteristic polynomial:

$$(10 - \lambda)^2 - 25 = 0 \implies \lambda_1 = 15, \lambda_2 = 5$$

The corresponding eigenvectors are:

$$\mathbf{v_1} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{v_2} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Therefore, the principal components are:

$$PC_1 = \overline{X}v_1 = \begin{bmatrix} -\frac{2}{\sqrt{2}} \\ -\frac{2}{\sqrt{2}} \\ -\frac{2}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ \frac{4}{\sqrt{2}} \end{bmatrix}, \qquad PC_2 = \overline{X}v_2 = \begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 0 \\ \frac{2}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$$